(C-019-DOT) DC-DC Phase-Shift Full-Bridge Buck Converter (DOT247)

PLECS Demo Circuit
Simulation Example

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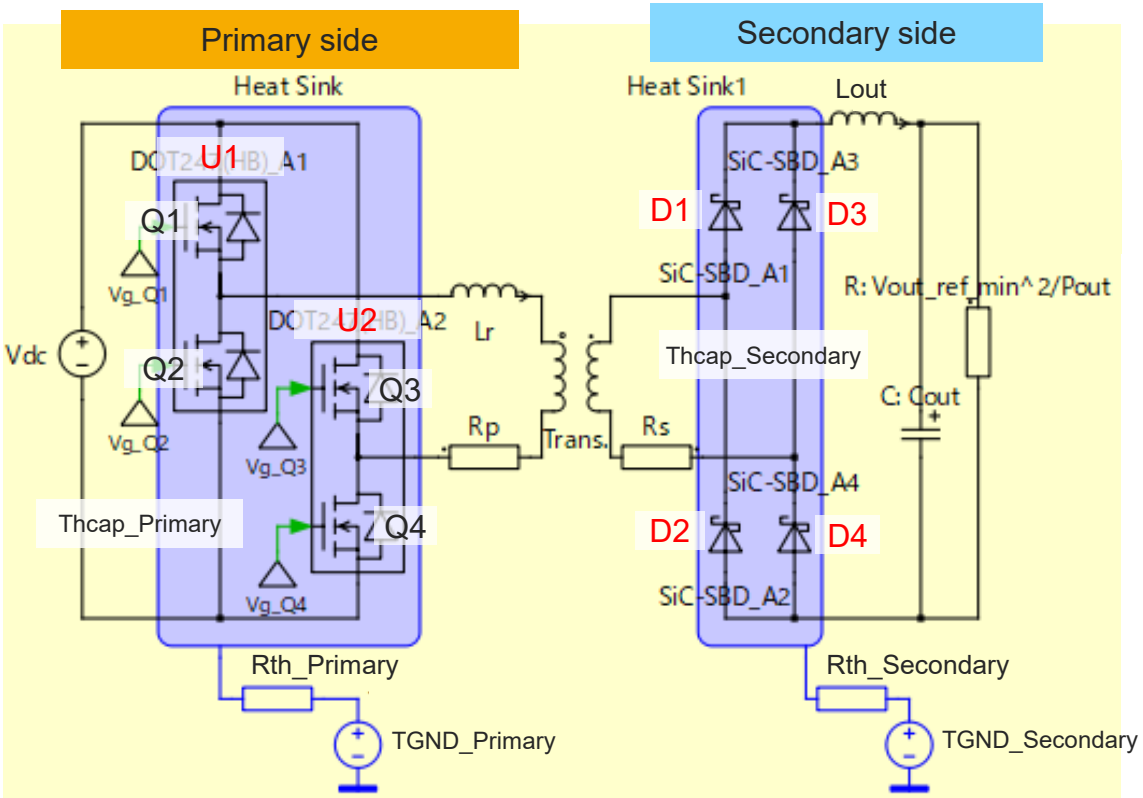
Simulation Parameters

	Name	Content	unit	Default Value
	Transformer	Np: Primary-turns Ns: Secondary-turns Lm: Magnetizing Inductance	turns turns H	20 18 1m
	Lr	Leakage Inductance	H	6.8u
	Lout	Output Inductance	H	330u
	Cout	Output Capacitance	F	47u
	Vc_init	Initial Voltage of Cout	V	300
	Rp	Parasitic Resistance (Primary)	ohm	5m
	Rs	Parasitic Resistance (Secondary)	ohm	5m
Primary	Thcap_Primary	Thermal Capacitance	J/K	0.1
Primary	Rth_Primary	Thermal Resistance	K/W	0.3
Primary	TGND_Primary	Thermal GND Temperature	℃	25
Secondary	Thcap_Secondary	Thermal Capacitance	J/K	0.1
Secondary	Rth_Secondary	Thermal Resistance	K/W	0.3
Secondary	TGND_Secondary	Thermal GND Temperature	℃	25

Name	Content	unit	Default Value
Test_time	Test time in simulation	s	0.15
fs	Switching Frequency	Hz	30k
Vin	Input Voltage	V	400
Vout_ref_min Vout_ref_max	Output Voltage	V	300 320
Pout	Output Power	W	5000
Rg_on*	Gate Resistance (Source)	Ω	15
Rg_off*	Gate Resistance (Sink)	Ω	15
DT	Dead Time	s	100n
T_init**	Initial Junction Temp.	℃	25

*Common for all MOSFETs　　**Common for all devices

Simulation Circuit



Power Devices

Name	Device Type	Part No.	Specification
U1, U2	SiC MOSFET module	SCZ4008DTA	750V/ 134A/ 8mΩ/ DOT247
D1～4	SiC SBD***	SCS320AG	650V/ 20A/ TO-220ACGE

*** SBD: Schottky Barrier Diode

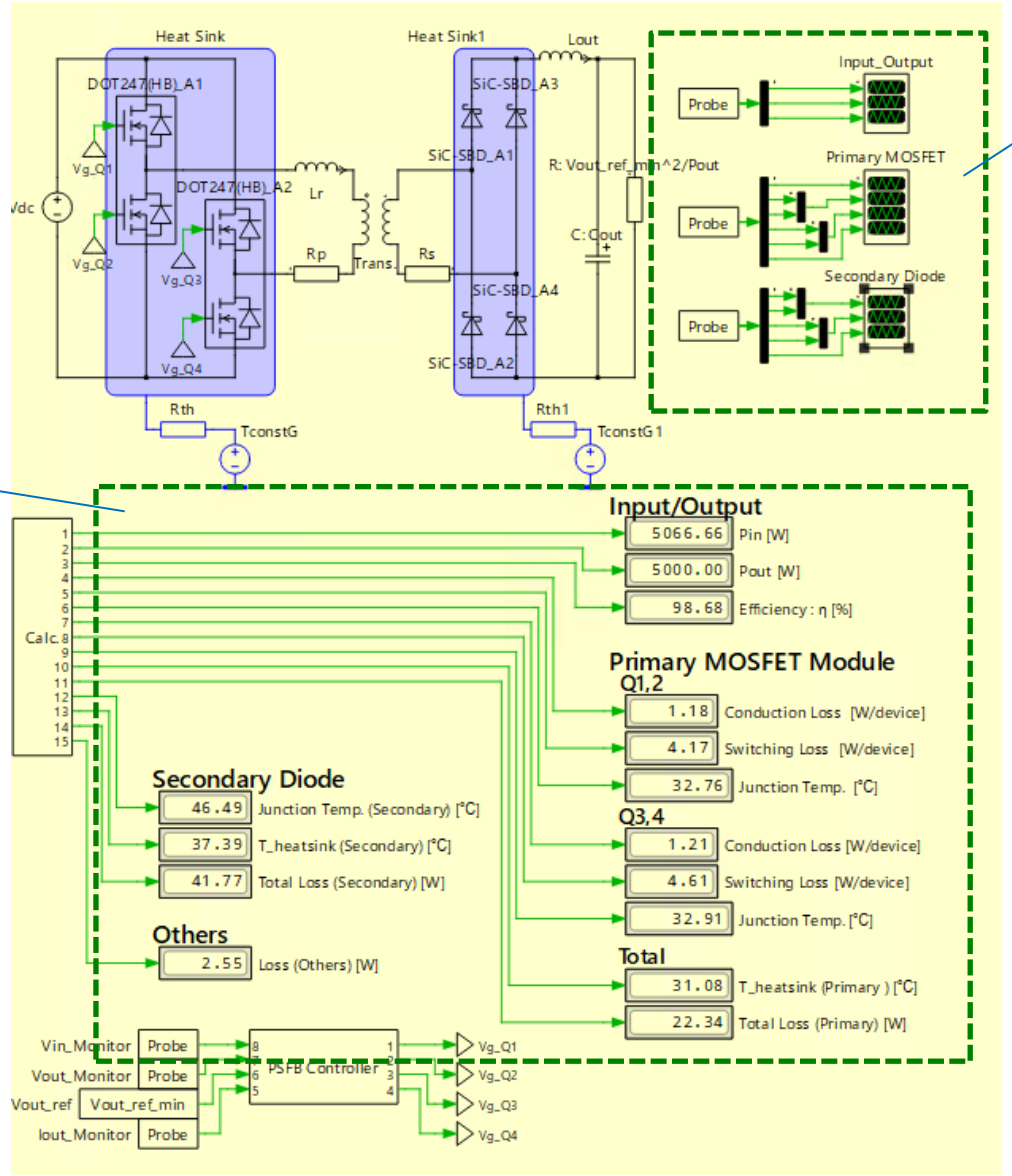
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Simulation Screen Overview

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Scope of the waveform

Results display



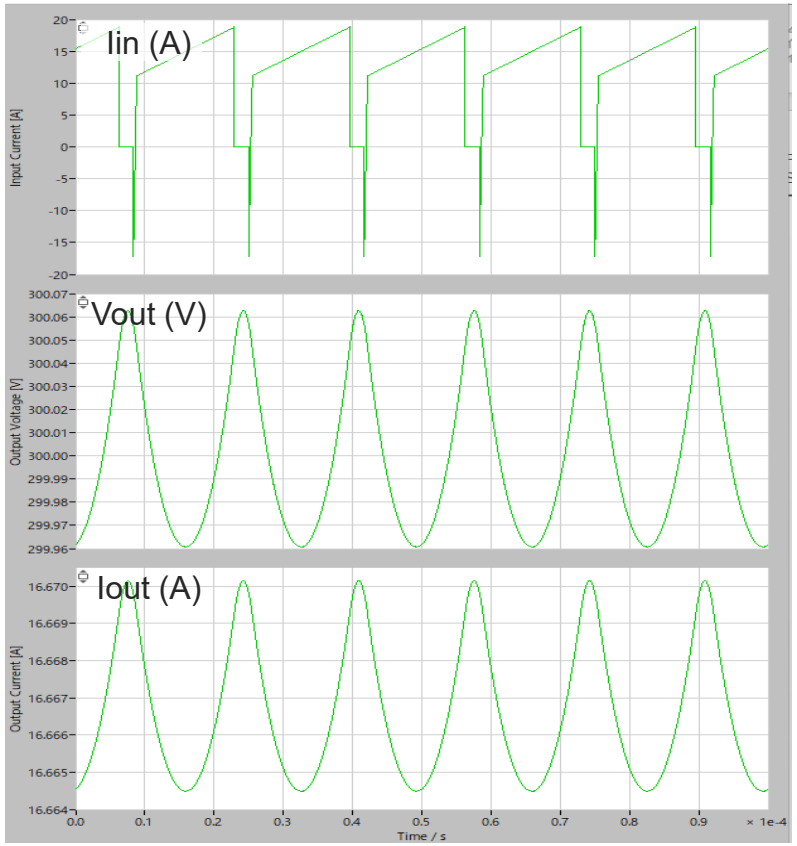
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Simulation Results

Simulation Mode: Steady State

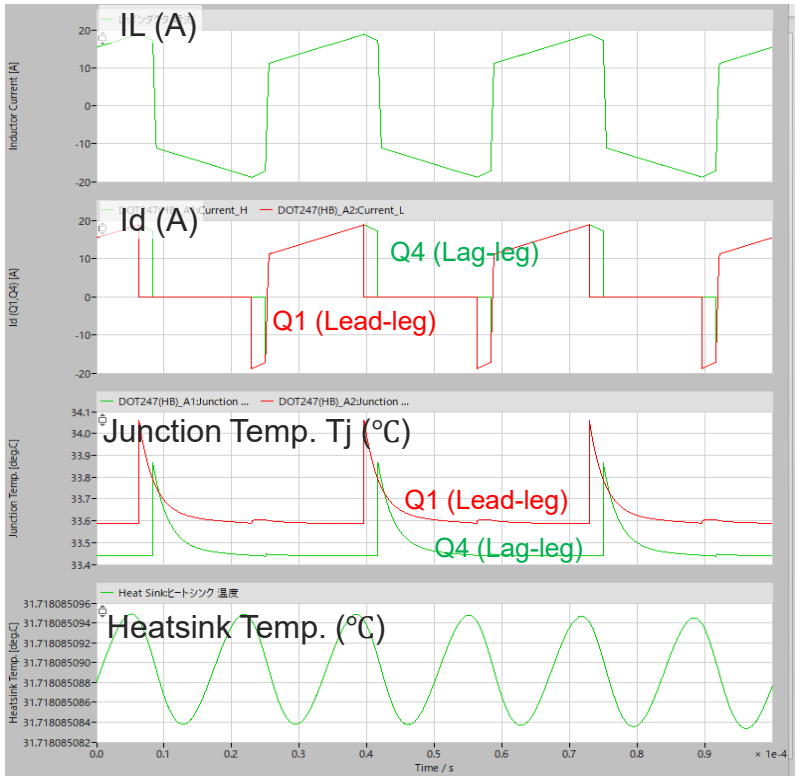
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Input and Output



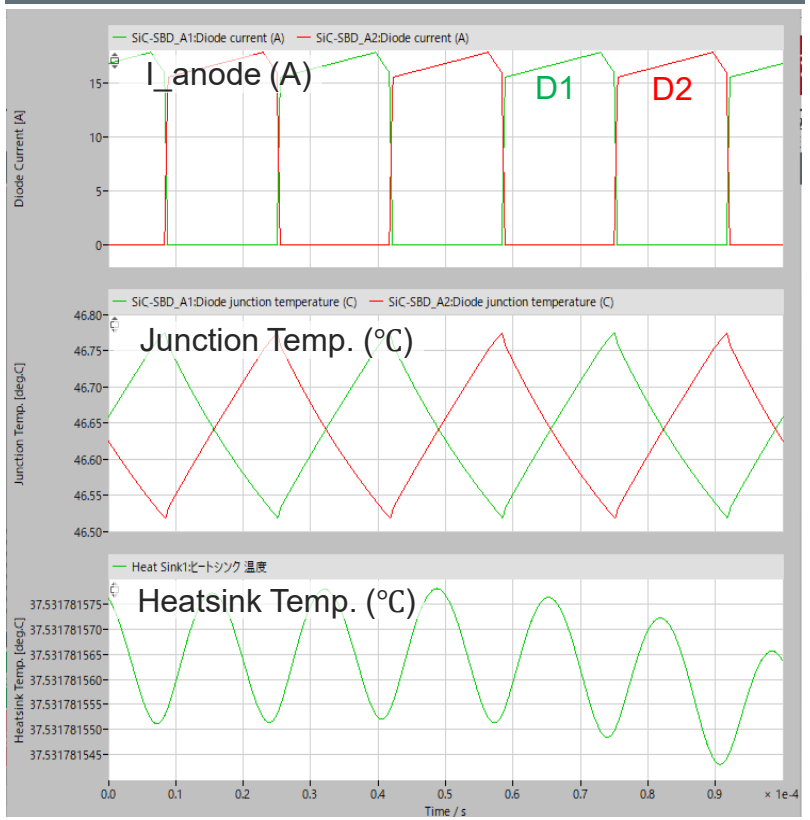
Contents	Results
Input Power : Pin	5.068 (kW)
Output Power: Pout	5.000 (kW)
Efficiency: η	98.66 (%)

Primary side



	Contents	Results
Q4 Lag-leg	Conduction Loss: Pcond (Q4)	1.22 (W/device)
	Switching Loss: Psw (Q4)	4.62 (W/device)
	Junction Temp. : Tj (Q4)	33.63 (°C)
Q1 Lead-leg	Conduction Loss: Pcond (Q1)	1.18 (W/device)
	Switching Loss: Psw (Q1)	4.18 (W/device)
	Junction Temp. : Tj (Q1)	33.47 (°C)
	Heatsink Temp.: T_hs (primary)	31.72 (°C)
	Total Loss: Ptot (primary)	22.39 (W)

Secondary side



Contents	Results
Junction Temp. Tj (secondary D1,D2)	46.64 (°C)
Heatsink Temp. T_hs (secondary)	37.53 (°C)
Total Loss: Ptot (secondary)	41.77 (W)

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To run this PLECS Reference Circuit, you must have the "PLECS" simulation software installed on your computer.

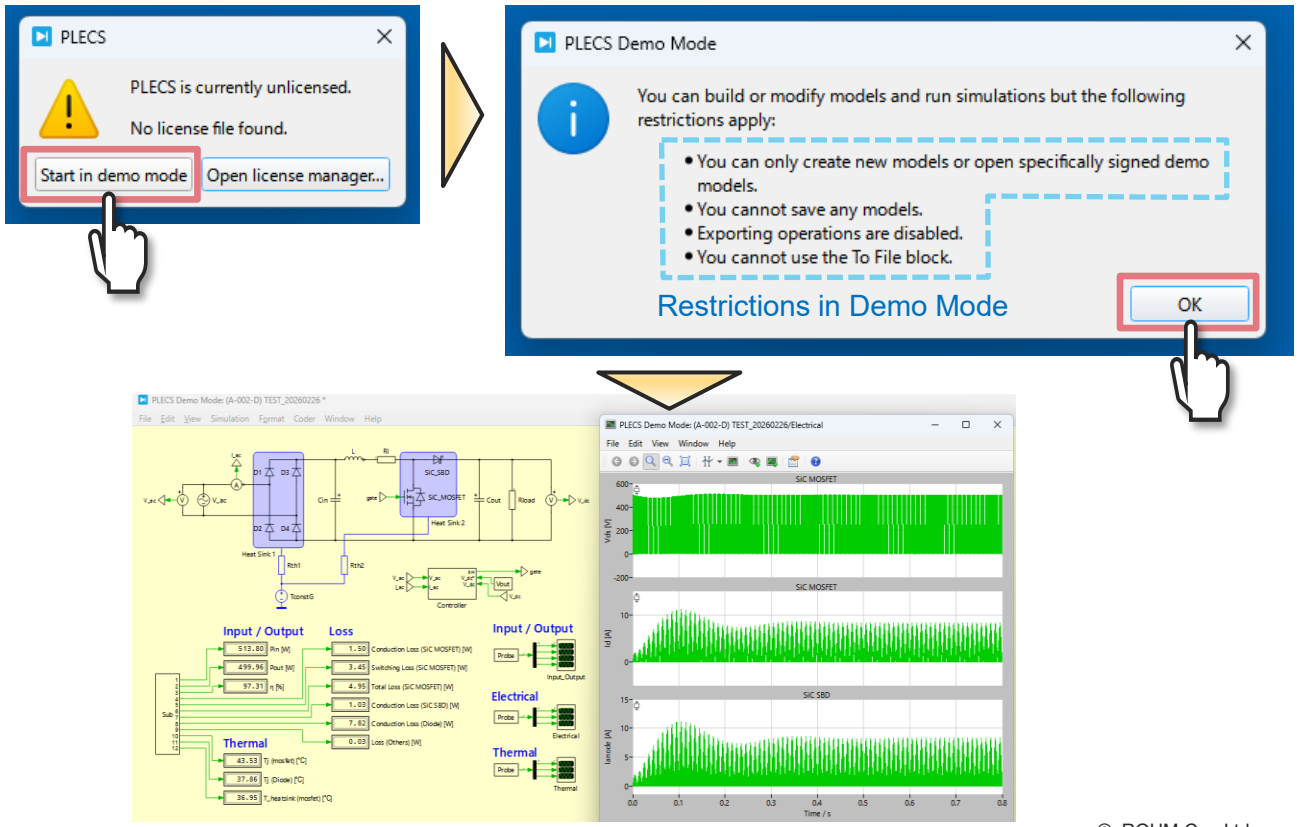
1. Obtaining the Software

If you do not have PLECS installed, please download the installer from the official website and complete the setup process.

https://www.plexim.com/download

2. Licensing (Using Demo Mode)

This reference circuit can be executed and viewed in "Demo Mode" even if you do not possess a paid license.

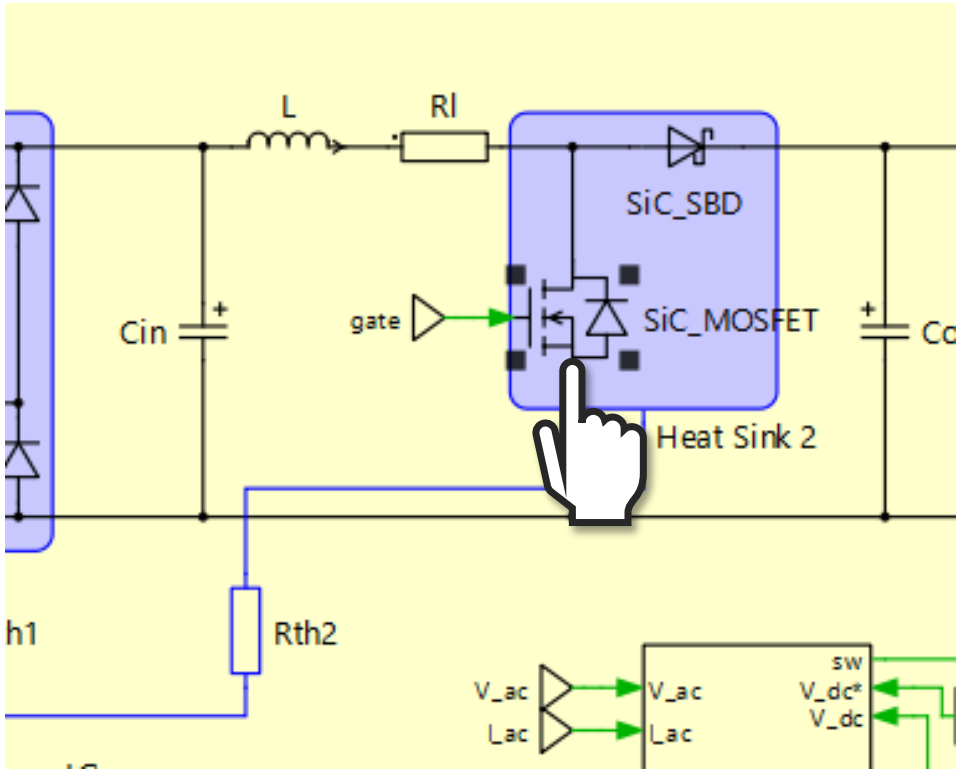


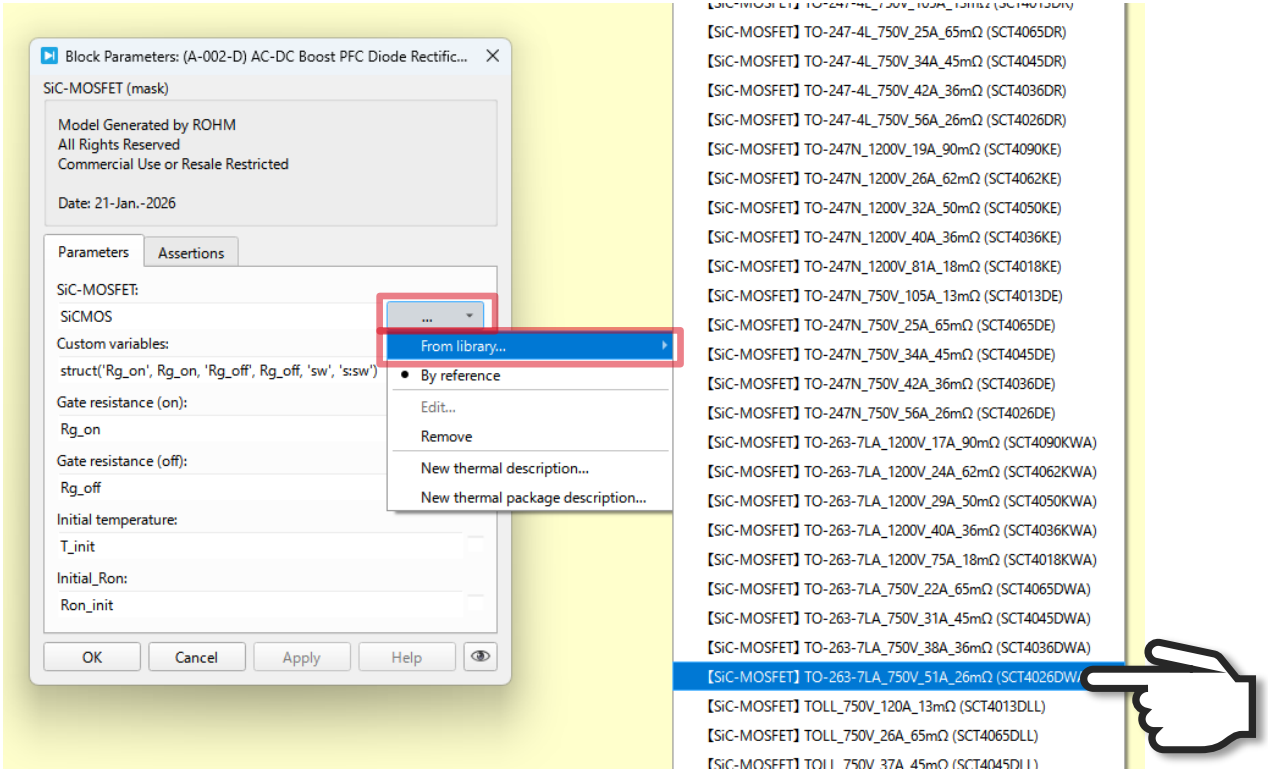
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How to change the devices

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Schematic window





Hover your mouse cursor over the device symbol that you want to change and double-click the left button of the mouse.

Click "..." > "From library..." to view the list of available devices and you can select a favorite device from these.

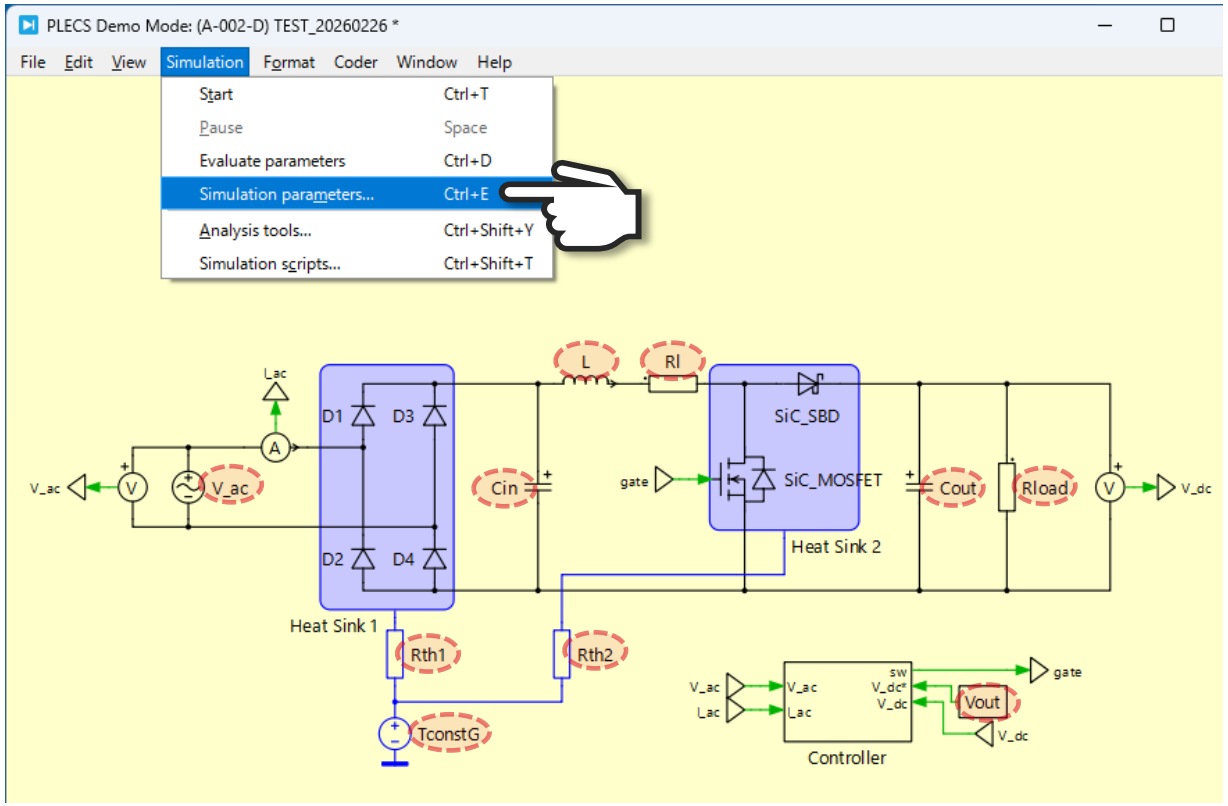
If the model you need is not in the list, please refer to the application note "How to Use PLECS Models".

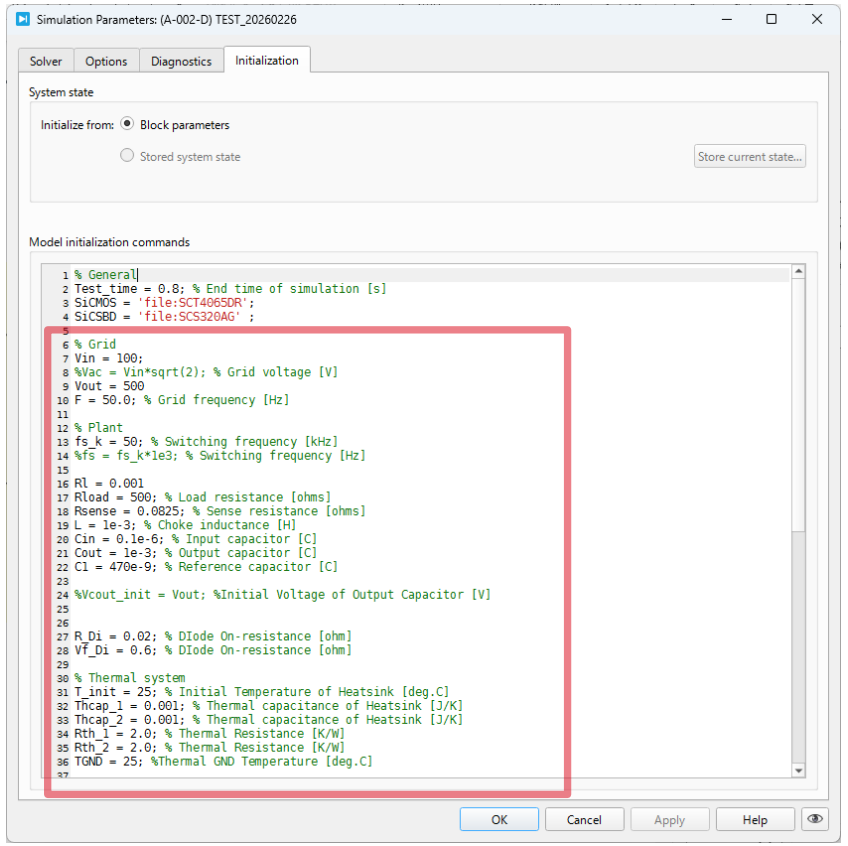
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How to change the simulation parameters

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Schematic window





All simulation parameters are parameterized. To modify them, go to the menu, select “Simulation parameters...” , and edit the values within "Model initialization commands."

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